

Surya Entrance Test [SET] 2027 - Session 02

Final Question Paper

Math-Sec A (3 Questions | +4 / -1)

Q1. Let

$$f(x) = \frac{x^3 - 3x + 2}{x^2 + 1}$$

The number of real values of x satisfying

$$f'(x) = 0$$

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q2. Let

$$A = \begin{pmatrix} 1 & a & 1 \\ a & 1 & a \\ 1 & a & 1 \end{pmatrix}.$$

If

$$|A| = 0,$$

then the sum of all real values of a is

- (A) -1
- (B) 0
- (C) 1
- (D) 2

Q3. Let

$$z = x + iy, \quad z \neq 0,$$

satisfy

$$\left| \frac{z-1}{z+1} \right| = 1.$$

- (A) Circle
- (B) Straight line
- (C) Ellipse
- (D) Parabola

Math-Sec B (4 Questions | +7 / -3)

Q4. Let

$$f(x) = x + \frac{1}{x}, \quad x \neq 0.$$

Consider the equation

$$f(x) = k,$$

where $k \in \mathbb{R}$. Which of the following statements are correct?

- (A) If $|k| < 2$, the equation has no real solution.
- (B) If $k=2$, the equation has exactly one distinct real solution.
- (C) If $k > 2$, the equation has exactly two distinct positive real solutions.
- (D) If $k < -2$, the equation has exactly two distinct negative real solutions.

Q5. Let

$$a, b, c$$

be non-zero vectors satisfying

$$|a| = |b| = |c| = 1$$

and

$$a + b + c = 0.$$

Which statements are correct?

(A) The angle between any two vectors is 120° .

(B) $a \cdot b = -\frac{1}{2}$

(C) $(a \times b) \cdot c = 0$.

(D) $|a + b| = 1$.

Q6. A fair coin is tossed four times. Let A: exactly two heads occur. B: the first toss is a head. C: the last toss is a tail. Which statements are correct?

(A) $P(A) = \frac{3}{8}$

(B) $P(A \cap B) = \frac{3}{16}$

(C) Events B and C are independent.

(D) $P(A | B) = \frac{3}{8}$

Q7. Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

For the matrix A, consider the following statements.

(A) The determinant of A is 3.

(B) The trace of A is 4.

(C) The eigenvalues of A are 1 and 3.

(D) The matrix A is invertible.

Q8. Let

$$R = \{(x, y) : 0 \leq x \leq 1, x^2 \leq y \leq 2x\}$$

Find 100 *times* the area of the region R .

[Numerical Answer Type — enter integer value]

Q9. Let

$$z = x + iy$$

satisfy

$$|z - 2| = |z + 2|$$

and

$$|z - (2 + 2i)| = 2.$$

Determine the maximum possible value of y . Enter the nearest integer.

[Numerical Answer Type — enter integer value]

Q10. A committee of 4 members is to be selected from 5 men and 4 women. If the committee must contain at least one woman, determine the total number of possible committees.

[Numerical Answer Type — enter integer value]

Math-Sec D (1 Questions | +20 / -7)

Q11.For real numbers

$$a, b, c$$

satisfying

$$a + b + c = 0,$$

and

$$a^2 + b^2 + c^2 = 6,$$

define

$$z_1 = a + bi, \quad z_2 = b + ci, \quad z_3 = c + ai.$$

Now define

$$F = \left| \prod_{k=1}^3 (z_k^4 + 4z_k^2 + 16) \right|.$$

Further define the triangle

$$A(a, b), \quad B(b, c), \quad C(c, a).$$

Suppose

$$P(x, y)$$

is a point inside triangle ABC satisfying

$$PA^2 + PB^2 + PC^2 = 18,$$

and

$$\vec{PA} + \vec{PB} + \vec{PC} = 0.$$

Now define

$$T = \int_0^\pi \ln \left(1 + \left| \sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2} \right| \right) dx.$$

Also define

$$S = \sum_{\text{cyc}} \frac{a^4}{(b-c)^2 + 1}.$$

Let

$$\lambda$$

be the greatest eigenvalue of

$$M = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}.$$

Finally define

$$\Omega = \frac{F}{\lambda^6} + 36T + S.$$

Determine exactly

$$\boxed{\Omega}.$$

(A) $\Omega = \frac{262144}{729} + 18\pi \ln 2 + 18$

(B) $\Omega = \frac{262143}{729} + 18\pi \ln 2 + 18$

(C) $\Omega = \frac{262144}{729} + 18\pi \ln 2 + 17$

(D) $\Omega = \frac{262144}{729} + 18\pi \ln 2 + 19$

Phy-Sec A (3 Questions | +4 / -1)

Q12. A particle moves on a frictionless track shaped so that its height above the horizontal is

$$h(x) = h_0 - \alpha x^2 + \beta x^4$$

where $\alpha, \beta > 0$. It is released from rest at $x=0$. Ignoring air resistance, which statement is necessarily true?

- (A) The speed is maximum exactly where the curvature of the track is maximum.
- (B) The speed becomes zero only once.
- (C) The acceleration is zero whenever the slope is zero.
- (D) The kinetic energy depends only on the change in height.

- Q13.** A conducting ring moves with constant speed into a region of uniform magnetic field perpendicular to its plane. The resistance of the ring increases uniformly with time due to heating. Which statement is correct?
- (A) The induced current always increases.
 - (B) The induced current always decreases.
 - (C) The induced current may increase initially and later decrease depending on the rate of change of resistance.
 - (D) The induced emf depends on resistance.
- Q14.** Two identical satellites move in circular orbits around Earth. Satellite A is at radius R . Satellite B is at radius $2R$. Which quantity has the same value for both?
- (A) Orbital speed and Orbital period
 - (B) Angular momentum and Orbital speed
 - (C) Mechanical energy per unit mass divided by orbital radius
 - (D) Orbital period and Angular momentum

Phy-Sec B (3 Questions | +7 / -3)

- Q15.** A uniform disc of mass M and radius R is free to rotate about its center. A light spring of spring constant k is wound around the rim without slipping. The free end is attached to a block of mass m resting on a frictionless horizontal surface. Initially the spring is unstretched and the system is at rest. The block is pulled slowly by distance x_0 and released. Neglect the spring's mass. Which of the following is/are correct?
- (A) The system executes SHM.
 - (B) The angular acceleration is maximum at equilibrium.
 - (C) Total mechanical energy remains constant.
 - (D) The angular velocity of the disc is maximum when the spring force is zero.

- Q16.** One mole of an ideal monoatomic gas undergoes the cyclic process 1. Isochoric heating 2. Isothermal expansion 3. Isobaric compression 4. Returns to the initial state Choose the correct statements.
- (A) Net work done over one cycle must always be positive.
 - (B) Internal energy after one complete cycle is unchanged.
 - (C) Heat absorbed equals work done over one cycle.
 - (D) Entropy change of the gas after one complete cycle is zero.
- Q17.** A photon strikes a stationary free electron. After collision, the scattered photon makes angle θ with the initial direction. Assume Compton scattering. Which statements are correct?
- (A) Wavelength of the scattered photon increases.
 - (B) Momentum of the electron is always perpendicular to the scattered photon.
 - (C) Photon energy decreases.
 - (D) Total momentum is conserved.

Phy-Sec C (2 Questions | +8 / -1)

- Q18.** A block of mass 2kg is attached to a spring of spring constant 200N/m on a smooth horizontal surface. The block is compressed by 0.30m and released from rest. Exactly at the equilibrium position, a perfectly inelastic collision occurs with another block of mass 2kg initially at rest on the same line. The two blocks stick together and continue moving while attached to the spring. Determine the maximum compression (in cm) of the spring after the collision (in cm).

[Numerical Answer Type — enter integer value]

Q19. A vacuum chamber contains a perfectly conducting horizontal rail of negligible resistance separated by 0.50 m. A conducting rod of mass 0.20 kg slides without friction on the rails. The rails are connected to a $2\ \Omega$ resistor and are placed in a uniform magnetic field of magnitude 4 T directed vertically downward. Initially, the rod is given a horizontal speed of 10 m/s. Assume: All electrical energy generated is converted into heat in the resistor. Exactly 20% of this heat is absorbed by 0.010 kg of an ideal monoatomic gas at constant volume. The gas initially has a temperature of 300 K. The heated gas emits thermal radiation that passes through a diffraction grating with 5000 lines/cm. The strongest emitted wavelength is determined using Wien's law

$$\lambda_{\max} T = 2.9 \times 10^{-3} \text{ m K}$$

This wavelength is then used to illuminate a metal surface having work function

$$\phi = 2.0 \text{ eV}.$$

Use Planck constant:

$$h = 6.63 \times 10^{-34} \text{ J s}$$

Speed of light:

$$c = 3.0 \times 10^8 \text{ m/s}$$

Electron charge:

$$e = 1.6 \times 10^{-19} \text{ C}$$

Gas constant:

$$R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$$

Assume the emitted wavelength is exactly equal to the wavelength incident on the metal. Find The maximum kinetic energy of the emitted photoelectrons (in eV). Enter the nearest integer.

Answer: _____

[Numerical Answer Type — enter integer value]

Phy-Sec D (2 Questions | +15 / -5)

Q20. A solid sphere of mass 4 kg and radius 0.20 m rolls without slipping from rest down a smooth curved track through a vertical height of 5 m. At the bottom, it immediately enters a rough horizontal surface of coefficient of kinetic friction 0.20. After travelling some distance, the sphere compresses a horizontal spring of spring constant 800 N/m. The rough surface continues beneath the spring. Assume: $g = 10 \text{ m/s}^2$. The sphere continues to roll without slipping throughout. The sphere travels 2.0 m on the rough surface before touching the spring. During compression, the sphere moves an additional 0.50 m before momentarily coming to rest. Find: The maximum spring compression (in cm). Round to the nearest integer (in cm)

[Numerical Answer Type — enter integer value]

Q21. A monochromatic laser beam of wavelength 500 nm is incident on a photoelectric cell. The light intensity is adjusted so that 2.0×10^{16} photons strike the cathode every second. The cathode has a work function of 2.0 eV. The emitted photoelectrons are collected and form a steady current through a 20Ω resistor. Assume: Every incident photon ejects exactly one electron. All emitted electrons reach the anode. Exactly 50% of the electrical power is converted into heat in 100 g of water. Initial water temperature is 20°C . Specific heat of water is $4200 \text{ J kg}^{-1} \text{K}^{-1}$. Heating continues for 1000 s. Use:

$$h = 6.63 \times 10^{-34} \text{ J s}$$

$$c = 3 \times 10^8 \text{ m/s}$$

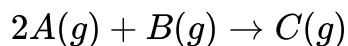
$$e = 1.6 \times 10^{-19} \text{ C}$$

Find: The increase in water temperature (in $^\circ \text{C}$). Enter the nearest integer.

[Numerical Answer Type — enter integer value]

Chem-Sec A (3 Questions | +4 / -1)

Q22. A gaseous reaction proceeds according to



The experimentally determined rate law is

$$r = k[A]^2[B]$$

Initially,

$$[A]_0 = 2.0 \text{ mol L}^{-1}, \quad [B]_0 = 1.0 \text{ mol L}^{-1}.$$

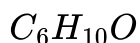
A catalyst is added that lowers the activation energy but does not alter the reaction mechanism. Simultaneously, the volume of the reaction vessel is suddenly doubled while maintaining the same temperature. Immediately after these changes, the reaction rate becomes what fraction of the original rate?

- (A) $\frac{1}{2}$
- (B) $\frac{1}{4}$
- (C) $\frac{1}{8}$
- (D) Depends on the value of the activation energy.

Q23. Consider the species NO_2^+ NO_2 NO_2^- NO . Which species contains the largest bond angle around the nitrogen atom?

- (A) NO_2^+
- (B) NO_2
- (C) NO_2^-
- (D) NO

Q24. A compound with molecular formula



shows the following observations: Gives a positive ferric chloride test. Does not react with Tollens' reagent. On bromination in aqueous medium, forms a white precipitate. On oxidation with alkaline $KMnO_4$, produces a compound that readily effervesces with sodium bicarbonate. The most probable functional group arrangement is

- (A) Phenol with an alkyl side chain
- (B) Aromatic aldehyde
- (C) Aromatic ether
- (D) Secondary alcohol attached to benzene

Chem-Sec B (4 Questions | +7 / --3)

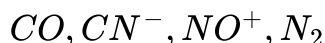
Q25. A galvanic cell is constructed using the half-cells



At 298 K, the concentration of Cu^{2+} is gradually decreased while the concentration of Zn^{2+} remains unchanged. Which of the following statements is/are correct?

- (A) The cell potential decreases.
- (B) The Gibbs free energy change becomes less negative.
- (C) The equilibrium constant of the cell reaction changes.
- (D) The driving force for electron flow decreases.

Q26. For the following species,



Identify the correct statement(s).

- (A) All are isoelectronic.
- (B) All possess bond order equal to 3.
- (C) All can act as ligands toward transition metals.
- (D) All are diamagnetic.

Q27. An unknown aromatic compound undergoes the following sequence: 1. Nitration 2. Reduction using Sn/HCl 3. Diazotization 4. Heating the diazonium salt with water Which statements are correct?

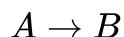
- (A) The final product contains a phenolic –OH group.
- (B) Nitrogen gas is evolved during the last step.
- (C) The reduction step converts a nitro group into an amino group.
- (D) The final product necessarily contains two hydroxyl groups.

Q28. A saturated aqueous solution of AgCl at 25°C is in equilibrium with excess solid AgCl. The following operations are performed independently. I. Addition of solid NaCl II. Addition of aqueous NH₃ (excess) III. Addition of distilled water IV. Increasing the temperature, assuming the dissolution of AgCl is endothermic. Which of the following statements are correct?

- (A) In operation I, the molar solubility of AgCl decreases.
- (B) In operation II, the molar solubility of AgCl increases.
- (C) In operation III, the value of K_{sp} increases.
- (D) In operation IV, the solubility of AgCl increases.

Chem-Sec C (4 Questions | +8 / -1)

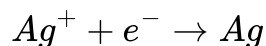
Q29. For the first-order reaction



the half-life is 20 minutes. Initially, 80 g of A is present. After 100 minutes, determine the mass (in g) of A remaining.

[Numerical Answer Type — enter integer value]

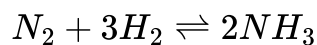
Q30. A current of 4.825 A is passed through aqueous AgNO₃ for 20 minutes. Assume 100% current efficiency.



Given $F = 96500 \text{ C mol}^{-1}$ Molar mass of Ag = 108 g/mol Determine the mass of silver deposited (in g).

[Numerical Answer Type — enter integer value]

Q31.For the reaction



the equilibrium constant is

$$K_c = 16$$

Initially,

$$[N_2] = 1M$$

$$[H_2] = 4M$$

$$[NH_3] = 0$$

At equilibrium, let the concentration of ammonia formed be $2x$. Determine the nearest integer value of $100x$.

[Numerical Answer Type — enter integer value]

Q32.A hydrocarbon contains 90% carbon and 10% hydrogen by mass. Its vapour density is 20. Determine the number of hydrogen atoms present in one molecule.

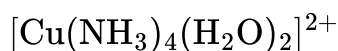
[Numerical Answer Type — enter integer value]

Chem-Sec D (1 Questions | +15 / -1)

Q33. A laboratory prepares an unknown coordination compound X through the following sequence.

Step I: Exactly 0.10 mol of pure copper reacts completely with concentrated nitric acid to produce $\text{Cu}(\text{NO}_3)_2$. The solution is evaporated to dryness and the residue is dissolved in water.

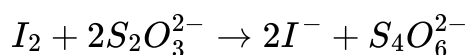
Step II: Excess aqueous ammonia is added until the deep-blue complex



is completely formed. No copper is lost during this step. *Step III:* The resulting solution is electrolysed using inert platinum electrodes until exactly 50% of the copper ions are deposited as metallic copper. Assume: Current efficiency = 100% $F = 96500 \text{ C mol}^{-1}$ *Step IV:* The remaining solution is treated with excess sodium hydroxide, producing copper(II) hydroxide, which is heated strongly to form CuO . The CuO is then completely reduced by hydrogen gas to metallic copper. *Step V:* The recovered copper is reacted with excess bromine to form CuBr_2 . The CuBr_2 is allowed to react with excess KI . The liberated iodine is titrated with



Reaction:



Find Determine the volume (in mL) of 0.50 M sodium thiosulfate required for complete titration. Round your answer to the nearest integer.

[Numerical Answer Type — enter integer value]